

# 1 Problem

It is an individual assignment.

Duration: 75 minutes.

Paper exam. Close book. No documents are allowed.

In this exam, students are required to develop a C++ console application for Football League Management.

The app will simulate the processes of entering match results and generating the league's ranking table.

## Task of inputting the match result:

Students are tasked to implement a feature that allows the league staff to input a match result, including the name of the home and away teams, as well as the number of goals scored by each team. For example: NEW vs CHEL 4-1. All match results must be stored in a text file for subsequent tasks.

## Task of outputting the ranking table:

The ranking table should be in the tabular format, including the following columns: #, team, MP, W, D, L, GF, GA, GD, Pts. The top k teams will qualify for an international league. The value of k may vary over seasons but remains constant in a specific season.

- #: rank number
- Team: name of team
- MP: number of matches
- W: number of winning matches
- D: number of draw matches
- L: number of losing matches
- GF: Goals For
- GA: Goals Against
- GD: Goals Difference,  $GF - GA$
- Pts: total points.

The current system for awarding points is as follows: 3 points for a win, 1 point for a draw, and no point for a loss.

The table must be sorted primarily by total points (in descending order), followed by GD (in descending order), and then by GF (in descending order).

But in some leagues, the ranking can be sorted by total points, followed by head-to-head results, and then by GD, and finally by GF.

Students must submit both a class diagram and the C++ source code.

The four principles of OOP must be applied in the exam.

Finally, students are free to use standard C++ libraries such as string, vector, set, map, and rand.

| Club                                                                                              | MP | W | D | L | GF | GA | GD | Pts |
|---------------------------------------------------------------------------------------------------|----|---|---|---|----|----|----|-----|
| 1  Arsenal       | 13 | 9 | 3 | 1 | 27 | 10 | 17 | 30  |
| 2  Man City      | 13 | 9 | 2 | 2 | 33 | 13 | 20 | 29  |
| 3  Liverpool    | 13 | 8 | 4 | 1 | 28 | 11 | 17 | 28  |
| 4  Tottenham   | 12 | 8 | 2 | 2 | 24 | 15 | 9  | 26  |
| 5  Aston Villa | 12 | 8 | 1 | 3 | 29 | 17 | 12 | 25  |
| 6  Newcastle   | 13 | 7 | 2 | 4 | 31 | 14 | 17 | 23  |
| 7  Brighton    | 13 | 6 | 4 | 3 | 28 | 23 | 5  | 22  |
| 8  Man United  | 12 | 7 | 0 | 5 | 13 | 16 | -3 | 21  |
| 9  West Ham    | 13 | 6 | 2 | 5 | 23 | 23 | 0  | 20  |
| 10  Chelsea    | 13 | 4 | 4 | 5 | 22 | 20 | 2  | 16  |